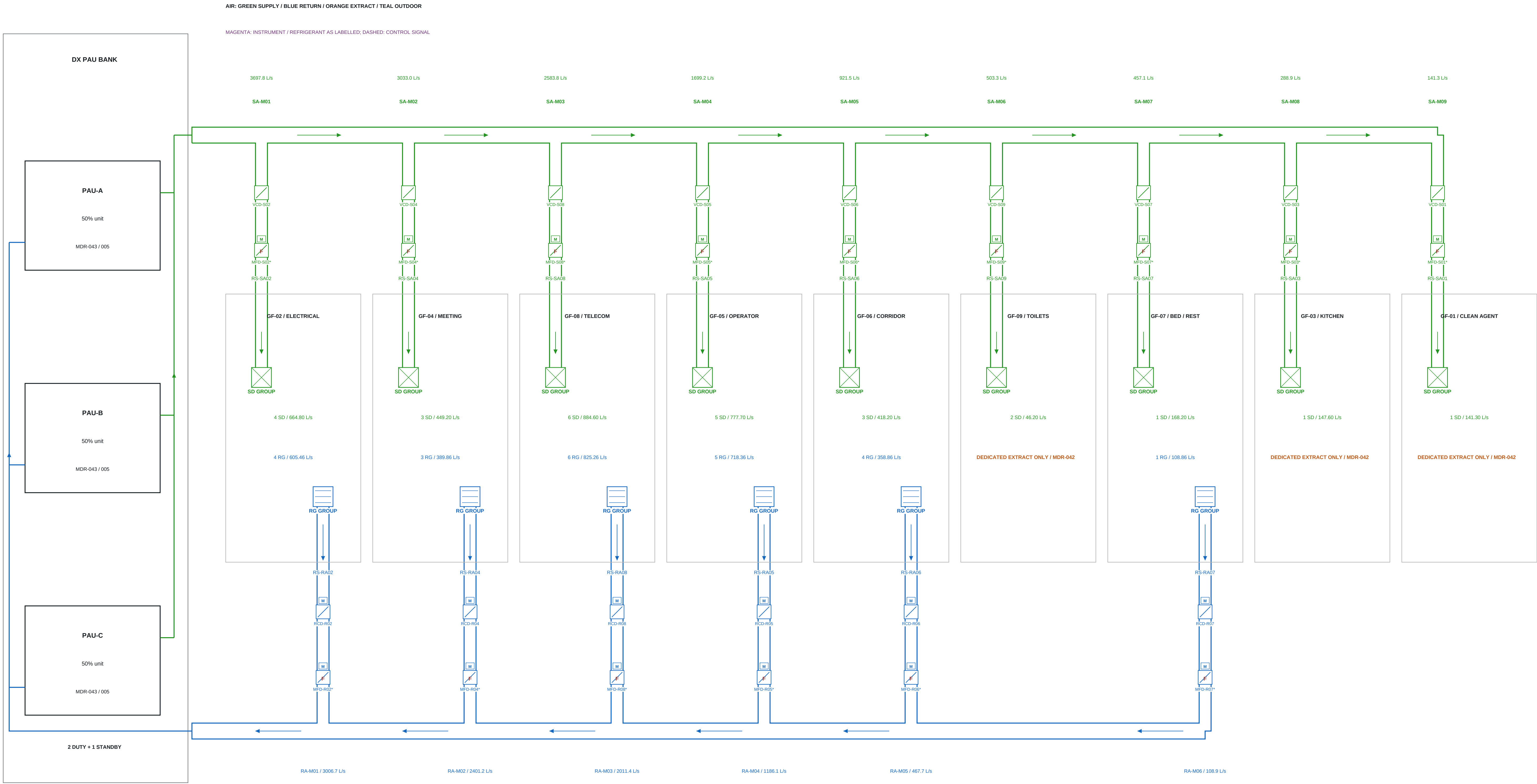


MAIN DUCT AND RETURN - WHOLE SYSTEM AIRFLOW METHOD



HOW THE METHOD WORKS

DX cooling takes place inside the PAUs. Refrigerant stays within the selected factory DX system; supply and return mains carry AIR. Common SUPPLY and RETURN mains stay on the ROOF. Nine supply drops and six return risers connect room branches to the shared headers; every PAU connects to the same header bank. Terminal groups and all instruments are expanded on the following sheets.

Supply 3697.80 L/s; assumed return 3006.68 L/s; minimum mass-balance makeup 691.12 L/s. DX cooling capacity / outside air / actual pressures remain unselected. Rooms with dedicated extract have no common return branch.

NTS functional air diagram: room order and sequential flow steps are illustrative; actual grouped roof takeoffs / coordinates on MDR-023. MFD* conditional on approved fire boundaries; RCD pressure modulation is a proposal. All room flows retained. Main sizes: MDR-050.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

WHOLE SYSTEM AIRFLOW METHOD

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-040

SCALE: NTS AT A1

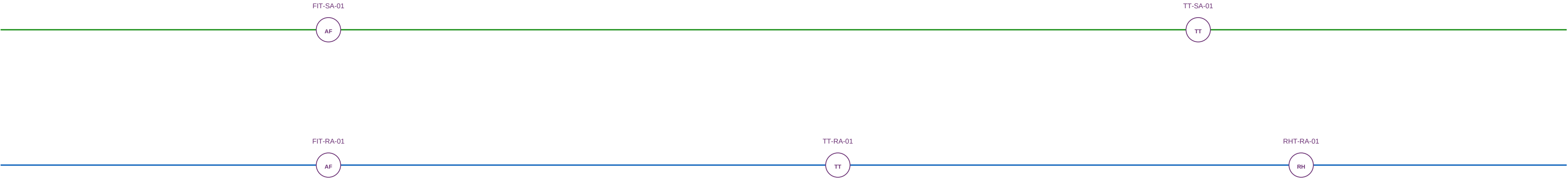
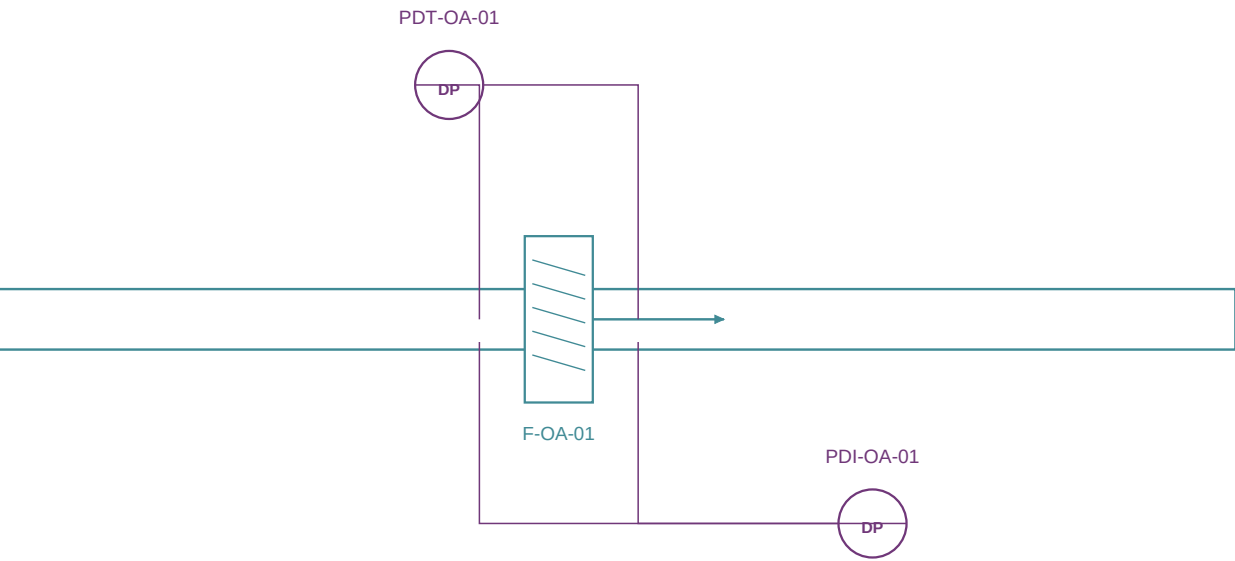
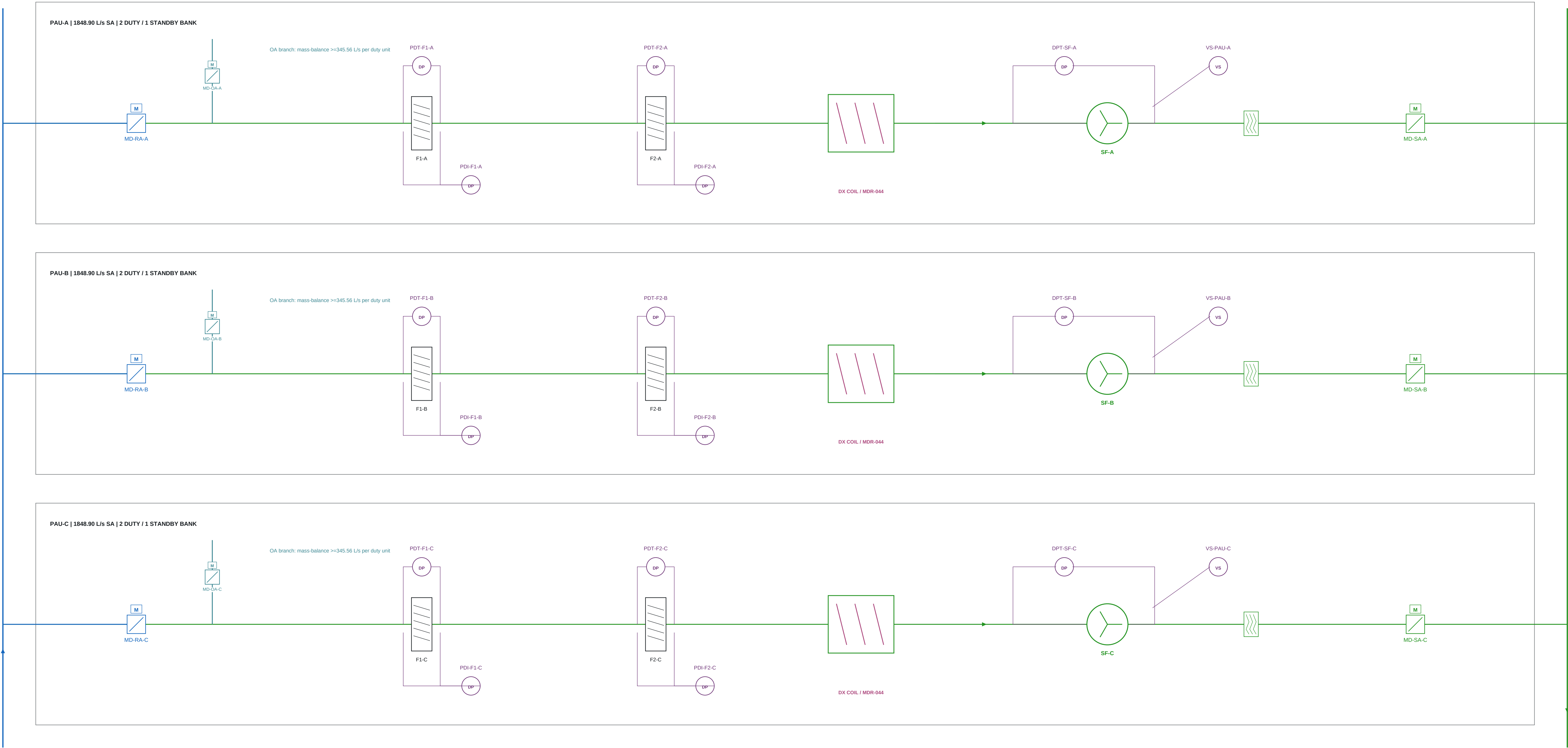
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08 OCT 2026

SHEET 40 OF 50

COMMON RA

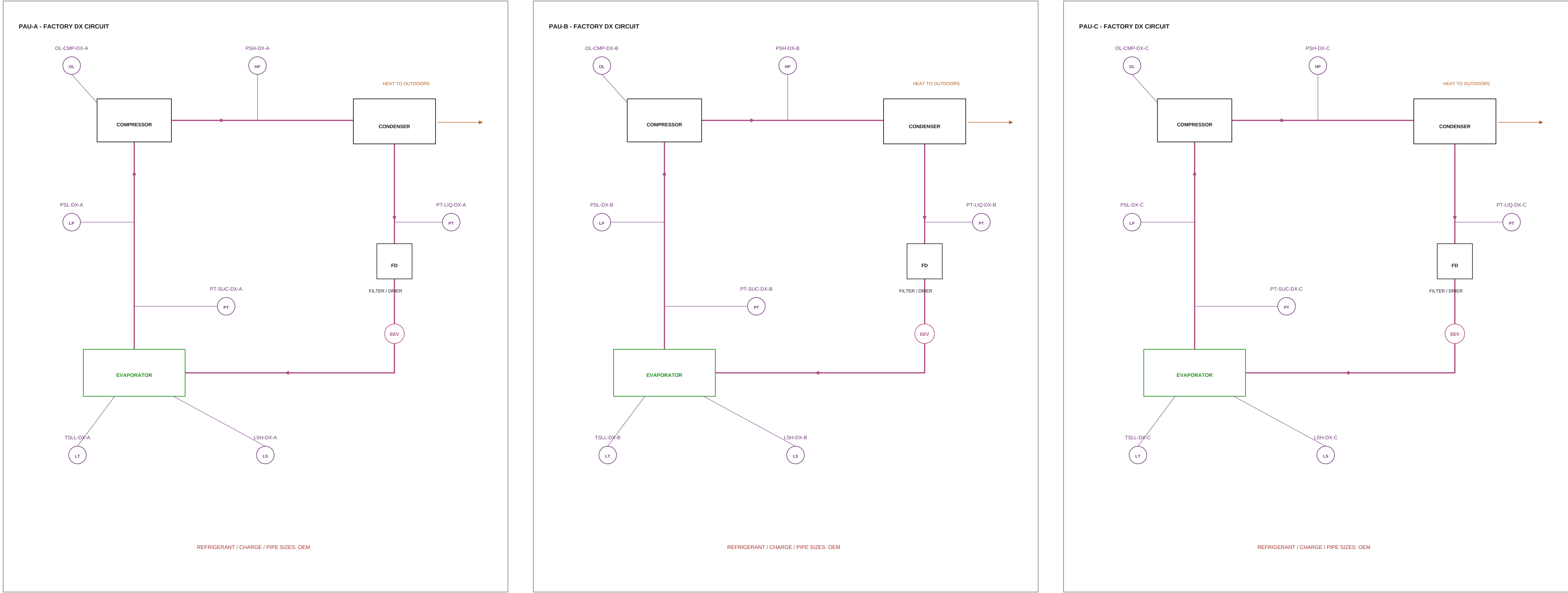
COMMON SA



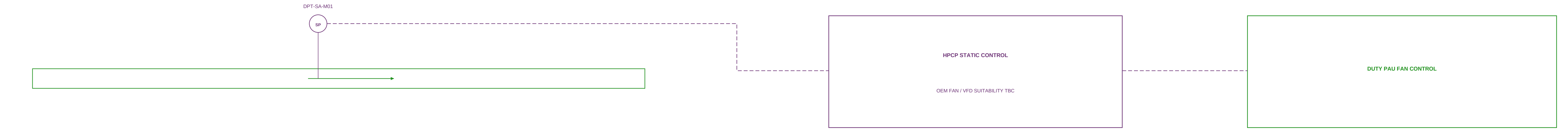
Standby PAU supply / return / OA paths close and prove isolated; duty paths open and prove before start. Common airflow stations require OEM straight runs. Unit nozzle routing, fan ESP, filters, cooling capacity and outdoor air remain unselected.

ADMINISTRATION HVAC / DX PAU OPTION COMMON SUPPLY MAIN + COMMON RETURN MAIN FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION	DRAWING TITLE DX PAU BANK, COMMON HEADERS AND AIR-SIDE INSTRUMENTS ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING	ADM-HVAC-MDR-043 SCALE: NTS AT A1	DRAWING REVISION 02 08 OCT 2026 SHEET 43 OF 50
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MAIN DUCT AND RETURN - DX REFRIGERANT CYCLE AND PROPOSED OEM PROTECTION



OPTIONAL COMMON-MAIN STATIC CONTROL / NEW PROPOSAL



22 new DX protection / monitoring candidates are separate from the 85 retained air-side instruments. Factory compressor / condenser / evaporator control governs. The shown circuit explains DX operation; receiver, accumulator, refrigerant, valve, oil return, relief and electrical arrangements follow the selected OEM circuit.

No field refrigerant network to the rooms is proposed. Cooling capacity, refrigerant pressure-temperature ranges and compressor quantity/staging are TBC. Do not modify factory safety wiring from this diagram.

ADMINISTRATION HVAC / DX PAU OPTION COMMON SUPPLY MAIN + COMMON RETURN MAIN FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION	DRAWING TITLE DX REFRIGERANT CYCLE AND PROPOSED OEM PROTECTION ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING	ADM-HVAC-MDR-044 SCALE: NTS AT A1	DRAWING REVISION 02 08 OCT 2026 SHEET 44 OF 50
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